Peer Review: "Selecting Bayesian VAR Shrinkage Hyperparameters by Forecast Loss vs. Marginal Data Density" (repository: laurentflorin/paper_hyperparameter_optimization)

TL;DR

The repository is a well-engineered but **incomplete-as-a-paper** research codebase asking a genuinely interesting econometric question — does selecting Bayesian VAR shrinkage hyperparameters by pseudo-out-of-sample forecast RMSE beat the principled marginal-data-density (MDD) rule of Giannone, Lenza & Primiceri (2015)? — but as presented it is **not yet acceptable for a peer-reviewed venue** because there is no readable manuscript, no reported results, and no statistical inference on forecast differences.

The most serious scientific threats are: (1) **no significance testing** (no Diebold–Mariano or equivalent), so any RMSE ranking is uninterpretable; (2) a **single-run, tiny-budget optimizer design** (Mango with ~5 init + 15 iterations, `n_eval=3` inner origins) that confounds "selection rule" with "optimizer noise"; and (3) **self-disclosed correctness/reproducibility defects** — a degenerate `1e10` RMSE penalty from an MBFVAR dimension mismatch, and an un-seedable upstream RNG that prevents bit-for-bit reproduction.

Recommendation: **Major revision.** As a reproducible code artifact it is above average for academic econometrics; as a scientific paper the empirical claims are not yet substantiated. The remedy is largely additive: add Diebold–Mariano/Wilcoxon tests, multiple seeds with variance reporting, an equal-budget/equal-care objective comparison, and a written results section.

Key Findings

This is an econometrics study, not a generic ML-HPO benchmark. The object being tuned is the shrinkage/prior hyperparameter vector of a Bayesian VAR (the Giannone–Lenza–Primiceri quarterly BVAR) and of a Schorfheide–Song mixed-frequency VAR (MFVAR). The two "HPO methods" compared are **not** grid vs. random vs. TPE; they are two *objective functions* for the same Bayesian optimizer (Mango): marginal data density (MDD) vs. pseudo-out-of-sample RMSE. The work must therefore be judged against the BVAR forecasting literature, not the AutoML benchmark literature. [github](#) ↗

The research question is valid and well-motivated. Whether to tune priors by marginal likelihood or by out-of-sample loss is a live debate: Bańbura, Giannone & Reichlin (2010) tune shrinkage to in-sample fit; Giannone, Lenza & Primiceri (2015) maximize the marginal likelihood in a hierarchical framework; Carriero, Clark & Marcellino (2015) study specification choices empirically. A controlled MDD-vs-loss comparison is a legitimate contribution in principle. [Hedibert](#) ↗

The experimental scaffolding is unusually careful for an academic repo: four "selection scopes" (`pooled` / `horizon` / `variable` / `variable_horizon`), configurable selection schedules (`once` / `per_origin` / `annual_quarterly` / every-`N`), nested inner/outer real-time evaluation, configuration-hash-based resume logic, a CI-tested synthetic ridge path with no optional dependencies, and Slurm/Euler batch support with BLAS/OpenMP thread pinning.

The scientific reporting layer is missing. There is no `.tex`/PDF/manuscript file anywhere in the top-level tree; "results" exist only as CSV/PNG artifacts produced by the comparison scripts (`compare_forecasts.py`, `compare_glp_forecasts.py`, `compare_scope_study.py`). No hypotheses are operationalized as testable claims with effect sizes, and — based on everything visible — **no statistical significance testing is advertised or (apparently) implemented.**

Reproducibility is partly engineered but has a known hole. Dependencies are pinned (`requirements.lock`, `.python-version`, MBFVAR pinned to commit `5b06f93…`), but MBFVAR "uses `numpy.random.default_rng()` internally without a user-exposed seed path," so exact replication "may require patching the upstream package." No `LICENSE` file is visible. [github ↗]

A correctness bug is self-disclosed. For the MFVAR RMSE optimizer, "smaller quarterly subsets such as `GDP` alone trigger an upstream `MBFVAR` forecast dimension mismatch and degenerate to the fixed `1e10` penalty that shows up as the optimizer's best score." A failure sentinel being recorded as a *best* score is a silent failure mode that can corrupt selected hyperparameters and all downstream forecasts.

Details

What the research does (reconstructed)

Question (verbatim from README): "does selecting hyperparameters by forecast loss produce better real-time macroeconomic forecasts than the principled Bayesian (MDD) rule?" [github ↗]

Two selection strategies: (1) *Native/MDD* — maximize the marginal data density as in Giannone, Lenza & Primiceri (2015); (2) *Forecast-loss* — minimize pseudo-out-of-sample RMSE on inner validation origins.

Models/applications: (i) a synthetic 3-variable ridge VAR (CI / no-deps path); (ii) the GLP quarterly BVAR in three nested sizes (small: GDP/DEFL/FFR; medium: +CONS/INV/HOURS/WAGE; large: +14 more series); (iii) the Schorfheide–Song 11-variable mixed-frequency VAR (GDP, INVFIX, GOV, UNR, HRS, CPI, IP, PCE, FF, TB, SP500). [github ↗] [github ↗]

Optimizer: Mango (ARM's parallel Bayesian-optimization library — a GP-based multi-armed-bandit optimizer), used both to maximize MDD and to minimize RMSE. Two RMSE variants: last-`n_eval` rolling origins vs. a random fixed sample of valid origins. RMSE runs are horizon-specific by default (1/2/4/8 quarters). The GLP "paper" strategy integrates the hyperparameters out via a random-walk Metropolis step — consistent with Giannone, Lenza & Primiceri (2015), who treat hyperparameters "as additional parameters, in the spirit of hierarchical modeling," placing hyperpriors on all of them and estimating them via a random-walk MH step. [github ↗]

Data: real US macro data via FRED/ALFRED real-time vintages (SS: origins 1997-07 to 2010-01, evaluation vintage 2012-01; GLP: recursive 2000-03-31 to 2019-12-31, scoring vintage 2023-01-01), with a Stooq S&P 500 fallback when the ALFRED SP500 vintage is unavailable. [github ↗]

Budget (from README examples): MDD/RMSE optimization with roughly 5 init points + 15 iterations, `nsim=1000`, inner `n_eval=3` evaluation origins ("a compromise between runtime and objective stability"). [github ↗]

Metrics/aggregation: per origin/variable/horizon point forecasts, intervals, actuals, and errors; RMSE and relative-RMSE tables vs. the paper hyperparameters; hyperparameter-path figures. Forecast errors are computed on quarter-on-quarter log growth (in %) for GDP, INVFIX, GOV, HRS, CPI, IP, PCE, SP500, and on quarterly-average levels for UNR, FF, TB.

Is the methodology sound? Assessment by dimension

Research question & hypotheses. Valid and clearly posed at a high level, but not operationalized into falsifiable hypotheses with pre-specified effect sizes, target variables, and horizons. "Better forecasts" is under-defined: better on which variables, at which horizons, and by a margin large enough to matter?

Experimental design / fairness of comparison. The central comparison (MDD objective vs. RMSE objective) is fair only if both objectives receive the *same optimizer budget and the same evaluation care*. The RMSE objective is estimated on only `n_eval=3` inner origins — a very noisy estimate of generalization loss — whereas the MDD objective is a smooth, near-closed-form quantity. This asymmetry is a confound that could bias the comparison in either direction. There is also an inherent **objective-mismatch advantage**: the RMSE-selected model is tuned on the *same loss* (RMSE) later used to evaluate it, structurally favoring it unless inner/outer separation is airtight (see leakage below).

Statistical rigor. This is the single biggest gap. There is no evidence of Diebold–Mariano tests (the field standard for comparing the predictive accuracy of two forecasts; Diebold & Mariano 1995), nor of the Harvey–Leybourne–Newbold small-sample correction for multi-step horizons, nor of Wilcoxon signed-rank / Friedman+Nemenyi comparisons across the variable×horizon grid (Demšar 2006), nor any variance reporting across seeds. With ~50–80 quarterly origins and strongly autocorrelated multi-step forecast errors, raw RMSE differences are very likely within noise. A single RMSE per cell, with no interval or test, cannot support any accuracy claim.

Overfitting the validation set / leakage. Selecting hyperparameters by inner RMSE and then reporting outer RMSE is legitimate *only* under strict temporal nesting (inner origins strictly precede the outer evaluation point, using only vintage-available data). The repo advertises real-time vintages and an inner/outer split — the right instinct — but the actual guarantee lives in `docs/EXPERIMENT_DESIGN.md` and the source, which could not be verified here (see Caveats). Selection-based and adaptive overfitting to validation origins are documented HPO phenomena and should be quantified via the validation-vs-test optimism gap.

Search-space design. The MFVAR uses fixed paper hyperparameters `[0.09, 4.30, 1.0, 2.70, 4.30]` as the reference. The search space around these for the RMSE/MDD runs — its bounds and its log/linear scaling — is not documented in a verifiable way. Search-space choice frequently dominates optimizer outcomes, so it must be reported explicitly. [github](#) ↗

Benchmark selection & diversity. Two real applications (GLP, SS) plus one synthetic panel is thin for a general claim about "forecast-loss vs. MDD." Importantly, the standard ML HPO benchmark suites (HPOBench, YAHPO Gym, HPO-B, LCBench, NAS-Bench) are **inappropriate here** — this is a real-time macro-forecasting problem, not tabular ML — so their absence is *not* a defect, contrary to a generic HPO checklist. The right expansion is *more macro datasets/vintages, more countries, and more model sizes*, not ML tabular benchmarks.

Anytime vs. final performance; wall-clock vs. evaluations. Not reported. Because one MDD evaluation and one RMSE inner-evaluation differ greatly in per-iteration cost, a fair study must report results per equal *compute* budget, not merely per iteration count, and ideally show anytime (convergence) curves.

Reproducibility. Good pinning and config hashing; a genuine hole in RNG seeding via MBFVAR (`--seed-base`/`--optimization-random-seed` exist at the script layer, but the upstream package's internal RNG is not user-seedable). Hardware is documented for the Euler/Slurm environment. Raw outputs (`forecast_panel.csv`, `selected_hyperparameters.csv`, `failed_origins.csv`, `run_metadata.json`) are written per run. **Licensing is absent** — this should be added before release.

Code-quality risks to validity. The self-disclosed `1e10` degenerate-penalty bug is the most concerning: if a failed evaluation is recorded as the optimizer's *best* score, the selected hyperparameters and downstream forecasts are silently corrupted for affected configurations. Reliance on legacy scripts that "do not write configuration hashes or run-state files" creates a provenance gap between exploratory and canonical runs. [github ↗]

Situating in the literature

The study sits at the intersection of two literatures.

Priors/tuning for BVARs. Giannone, Lenza & Primiceri (2015), *Review of Economics and Statistics* 97(2):436–451, established hierarchical marginal-likelihood selection of shrinkage hyperparameters (random-walk MH over hyperparameters with hyperpriors). Bańbura, Giannone & Reichlin (2010), *Journal of Applied Econometrics* 25(1):71–92, instead tune shrinkage to a target in-sample fit (defined as the percentage deviation of the in-sample MSFE from the no-change forecast's MSFE). Carriero, Clark & Marcellino (2015), *Journal of Applied Econometrics* 30(1):46–73 ("Bayesian VARs: Specification Choices and Forecast Accuracy"; FRB Cleveland WP 1112, 2011) is the closest prior empirical work — they examine how tightness, lag length, multi-step method (direct/iterated/pseudo-iterated), and error-variance treatment affect BVAR forecast accuracy, concluding they "find very small losses (and sometimes even gains) from the adoption of specification choices that make BVAR modeling quick and easy." Schorfheide & Song (2015), *JBES* 33(3):366–380, provide the MFVAR.

Evaluation methodology. Diebold & Mariano (1995) and the Harvey–Leybourne–Newbold (1997) small-sample correction are the standard predictive-accuracy tests; Tashman (2000) and Bergmeir & Benítez on rolling-origin evaluation.

HPO framing (not benchmarks). Bergstra & Bengio (2012), *JMLR* 13:281–305 (random vs. grid search: "randomly chosen trials are more efficient for hyper-parameter optimization than trials on a grid"); Snoek, Larochelle & Adams (2012) on practical Bayesian optimization; Bischl et al. (2023, *WIREs Data Mining & Knowledge Discovery* 13(2):e1484) on HPO best practices and nested resampling; Probst, Boulesteix & Bischl (2019, *JMLR* 20(53)) on tunability; Demšar (2006, *JMLR* 7:1–30) on statistical comparison across many tasks; and the meta-overfitting literature (selection-based vs. adaptive overfitting of validation performance). The revision should explicitly cite and compare against Carriero–Clark–Marcellino and adopt Diebold–Mariano testing as its inferential backbone.

Novelty. A controlled comparison of MDD-objective vs. RMSE-objective selection *using the same Bayesian optimizer, in real time, across scopes and horizons* is a modestly novel and useful contribution if properly analyzed. It is incremental relative to Carriero–Clark–Marcellino but has a distinct angle — the *objective function used for tuning*, rather than model specification per se.

Reviewer Report

Summary of contribution

A reproducible Python framework that, for real-time US macro forecasting with GLP-style BVARs and the Schorfheide–Song MFVAR, compares two rules for choosing shrinkage hyperparameters — maximizing marginal data density vs. minimizing pseudo-out-of-sample RMSE — using the Mango Bayesian optimizer across four selection scopes, several schedules, and multiple horizons.

Overall assessment / recommendation

Major revision. The infrastructure is strong and largely reproducible, but the empirical contribution is not yet demonstrated: there is no manuscript, no reported/interpreted results, no statistical testing, and at least one self-disclosed bug that can corrupt results. None of these is fatal in principle; all are addressable with additive work.

Strengths

A clear, relevant research question grounded in a real methodological debate.

Careful real-time design (vintage-aware data, recursive origins, inner/outer nesting) that is exactly right for forecasting evaluation.

Substantial software quality: CI, pinned dependencies, config-hash resume, a no-deps synthetic path, Slurm/Euler support with thread pinning, and structured per-run output artifacts.

Transparent self-auditing (the repo ships `IMPLEMENTATION_AUDIT.md` and `CURRENT_STATE_AUDIT.md` and documents known limitations).

Major concerns (numbered)

No statistical inference on forecast differences. Any MDD-vs-RMSE ranking is uninterpretable without significance tests. *Threat:* headline conclusions could be pure noise. *Fix:* Diebold–Mariano (with HLN correction) per variable/horizon; Wilcoxon/Friedman+Nemenyi across the grid.

Single-run, tiny-budget optimizer design conflates selection rule with optimizer stochasticity. ~5+15 Mango iterations and `n_eval=3` inner origins are noisy. *Fix:* ≥10 seeds; report mean ± SE / bootstrap CIs; increase and sensitivity-test `n_eval`.

The `1e10` degenerate-penalty bug can silently corrupt selected hyperparameters. *Fix:* fail loudly on evaluation errors; audit all existing outputs for the sentinel value and rerun affected cells.

Unverifiable leakage safeguard. The inner/outer temporal nesting is claimed but not demonstrated. *Threat:* look-ahead bias would flatter the RMSE rule. *Fix:* document the nesting precisely and add a unit test that fails on any vintage look-ahead; report the validation→test optimism gap.

No equal-compute accounting. MDD and RMSE evaluations differ sharply in cost. *Fix:* report performance vs. wall-clock and vs. number of model estimations, with anytime curves.

No manuscript / no reported results. *Fix:* write the paper with tabulated, tested results and interpretation.

Minor concerns and presentation issues

No `LICENSE` file.

MFVAR search space and its scaling are undocumented.

Legacy scripts lack config hashes/run-state, creating provenance ambiguity.

Only three model families and two real datasets; consider more vintages/countries/sizes.

Reproducibility gap from the un-seedable MBFVAR RNG should be foregrounded, not buried in "Notes."

Concrete "what I would change"

Add experiments: multi-seed replication; equal-compute MDD-vs-RMSE comparison; ablation over `n_eval` (inner origins) and over search-space bounds; a "paper hyperparameters" baseline and a no-change/AIC-VAR baseline in every table (the code already supports `no_change`, `var_aic`).

Add statistical analysis: Diebold–Mariano (HLN-corrected) as primary; Wilcoxon/Friedman+Nemenyi with multiple-comparison correction across the grid; critical-difference-style ranked summaries across scopes.

Add reporting: per-variable/horizon RMSE with CIs; relative-RMSE with significance stars; validation→test optimism; anytime curves.

Fix code: eliminate the penalty-as-best-score path; expose/patch MBFVAR seeding; add a license and the committed raw result tables.

Questions to the authors

Is the inner-selection window guaranteed to precede the outer evaluation origin using only vintage-available data? Where is this enforced?

How many seeds/replications per configuration were actually run, and how stable are selected hyperparameters across seeds?
Are any results in `outputs/` affected by the `1e10` penalty, and were those filtered?
What are the exact search-space bounds and scaling for the MDD and RMSE optimizations?
Do you report/plan Diebold–Mariano or any significance testing? If implemented, where?
Why `n_eval=3`, and how sensitive are conclusions to it?
What is the intended target venue (econometrics vs. AutoML)? This determines the required baselines and framing.

Reproducibility checklist assessment

Code publicly available: **Yes.**
Pinned dependencies / environment: **Yes** (`requirements.lock`, `.python-version`, pinned MBFVAR commit).
Deterministic seeding: **Partial** — script-level seeds exist, but the upstream MBFVAR RNG is not user-seedable; bit-for-bit reproduction not guaranteed.
Hardware documented: **Yes** (Euler/Slurm, thread pinning).
Raw results available: **Partial/unclear** — per-run artifacts are written; committed final result tables not confirmed.
Statistical-analysis code: **Not evident.**
License: **Missing.**
Written manuscript with methods/results: **Missing.**

Caveats

Access limitation (important for calibrating this review): In this environment I could fully read only the repository README/landing page. The substantive internal documents central to several of my judgments — `IMPLEMENTATION_AUDIT.md`, `docs/EXPERIMENT_DESIGN.md`, `docs/CURRENT_STATE_AUDIT.md`, `docs/REPRODUCIBILITY.md`, `GLP_RMSE_REVIEW.md`, `README_GLP_METHODOLOGY.md` — and all source files under `src/` and `scripts/` **could not be retrieved** (both the fetch tool and a dedicated sub-investigation were blocked from GitHub blob/raw/API URLs for this unindexed repository). Consequently, statements about "no significance testing," "single-seed design," and the exact leakage safeguards are **inferences from the README's description of outputs and options**, and should be confirmed by cloning the repo and grepping for `diebold`/`dm_test`/`wilcoxon`/`scipy.stats` and inspecting the inner/outer split code. Where I could not verify, I have flagged it rather than asserting a defect as fact.
`IMPLEMENTATION_AUDIT.md` is described as containing severity-tagged findings with "status notes for resolved items," implying the authors already track issues; my Major-revision assessment would soften if that file shows the significance-testing and seeding gaps are already addressed and the `1e10` bug resolved.
This review evaluates the work as a *scientific paper*. As a *reproducible code artifact*, the repository is above the norm for academic econometrics; the gap is between "solid experiment engine" and "finished, statistically defensible study."